

Curriculum Vitae

Jinho Byun

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RESEARCH EXPERIENCE

Mar. 2025 – Present	Research Professor, Dept. of Energy Engineering, KENTECH
Mar. 2023 – Mar. 2025	Postdoctoral Researcher, Dept. of Energy Engineering, KENTECH
Aug. 2022 – Jan. 2023	R&D Engineer, Wonik IPS
Mar. 2022 – Jul. 2022	Postdoctoral Researcher, Dept. of Physics, Pusan National University (Advisor: Prof. Jaekwang Lee)

EDUCATION

Sep. 2017 – Feb. 2022	Ph.D., Physics, Pusan National University. Advisor: Prof. Jaekwang Lee. Thesis: “Emergent Phenomena at Polar Oxides and Their Applications.”
2012 – 2017	B.A., Physics, Pusan National University

HONORS AND AWARDS

2021	Outstanding Poster Award, Korean Physical Society (KPS) — “Interface engineered TER optimization in HfO ₂ -based FTJ”
2021	Excellent Academic Award, Pusan National University
2020	Humantech Paper Award, Samsung Electronics — “Anisotropic Evaporation of ZnO Observed by In-situ Cs-Corrected High Resolution TEM”
2019	Outstanding Poster Award, Korea Institute for Advanced Study (KIAS) — “First-principles study of the HfO ₂ -based ferroelectric tunnel junction”
2017	Outstanding Presentation Award, Korean Physical Society (KPS) — “Design of high-coercivity Fe ₁₆ -xAl _x N ₂ alloy”
2017 – 2020	Combined-course Scholarship (full tuition), Pusan National University
2012 – 2017	National Excellence Scholarship (Natural Sciences and Engineering), Korean Government (full tuition)

RESEARCH INTEREST

Methods: deep learning for microscopy, ab initio multislice image simulation, ab initio / machine-learning-force-field molecular dynamics, phonon properties, DFT

Materials systems: oxide surfaces and interfaces, perovskites, 2D materials

PUBLICATIONS

¶ equal/co-first contribution; * corresponding author.

Accepted / in press

1. Y. Xing¶, **J. Byun¶**, K. T. Kang, J. Hwang, J. Lee, W. S. Choi, S. H. Oh*. “Atomic-Scale Visualization of Vacancy-Ordered Intermediates and Pathway Selection in the Perovskite–Infinite-Layer Transformation.” *Journal of the American Chemical Society* (2026). DOI: 10.1021/jacs.6c08044. **Accepted; in production.**

Under review

2. K. Lee[¶], Y. Xing[¶], **J. Byun[¶]**, M. Jeong[¶], J. Seo, J. Lee, J. Bang, E. Lee*, S. H. Oh*. “Visualizing Oxygen Vacancy-Induced Dielectric Dead Layer at Metal-Oxide Interfaces by In-situ Multimodal Electron Microscopy.” *Science Advances* (under review, major revision).
3. J. Jeong[¶], **J. Byun[¶]**, X. Xu, A. Gruverman, J. Lee, S. H. Oh*. “Direct Visualization of Interfacial Defect Effects on Polarization Switching in BaTiO₃ Tunnel Junctions.” *Science Advances* (accepted, pending final technical revision).
4. Hojin Lee, Joonbong Lee, **J. Byun**, Sang-Hyeok Yang, Ayoung Cho, Sangwoo Lee, Dae Haa Ryu, Hyunbin Chung, Hee Seo Yun, Moon Seop Choi, Yesul Choi, Sungkyun Park, Min-Hyung Jung, Dongwon Lee, Yoon-Uk Heo, Hu Young Jeong, Y.-M. Kim*, J. Lee*, T. Choi*. “Interface-Engineered Hybrid Ferroelectric Tunnel Junctions Exhibiting Intrinsic Nonlinearity and Self-Rectification via Antiphase-Like Interlayer Formation.” *Advanced Functional Materials*, manuscript 7502704 (under review, second round).
5. Y. Xing[¶], **J. Byun[¶]**, J. Seo, Z. Wang, Y. Jin, S. Kim, J. Lee, S. H. Oh*. “Geometry-driven surface electron dichotomy via ordered subsurface defects.” *Nature Physics* (under review).

Peer-reviewed journal articles (published)

6. Y. Xing, I. Kim, K. T. Kang, J. Byun, W. S. Choi, J. Lee, S. H. Oh*. “Monitoring the formation of infinite-layer transition metal oxides through in situ atomic-resolution electron microscopy.” *Nature Chemistry* **17**, 66–73 (2025). [Author Correction: *Nat. Chem.* **17**, 777 (2025).]
7. J. Seo, H. Lee, K. Eom, J. Byun, T. Min, J. Lee, K. Lee, C.-B. Eom, S. H. Oh. “Field-induced modulation of two-dimensional electron gas at LaAlO₃/SrTiO₃ interface by polar distortion of LaAlO₃.” *Nature Communications* **15**, 5268 (2024).
8. J. Hwang, J. Byun, H. Kim, J. Lee. “High-mobility two-dimensional electron gas at the PbZr_{0.5}Ti_{0.5}O₃/BaSnO₃ heterostructure.” *J. Korean Phys. Soc.* **84**(1), 67–72 (2024).
9. Z. Wang[¶], **J. Byun[¶]**, Z. Wang, Y. Xing, J. Seo, J. Lee, S. H. Oh*. “Direct Observation of Atomic Step-Assisted Stabilization of Polar Surfaces.” *Advanced Materials* **35**(40), 2303051 (2023).
10. Z. Wang[¶], **J. Byun[¶]**, S. Lee, J. Seo, B. Park, J. Kim, H. Y. Jeong, J. Bang, J. Lee, S. H. Oh*. “Vacancy driven surface disorder catalyzes anisotropic evaporation of polar ZnO (0001) surface.” *Nature Communications* **13**, 5616 (2022).
11. M. Sheen, Y. Ko, D. Kim, J. Kim, J. Byun, K. Y. Yeon, D. Kim, J. Jung, J. Choi, R. Kim, J. Yoo, I. Kim, C. Joo, N. Hong, J. Lee, S. H. Jeon, S. H. Oh, J. Lee, N. Ahn*, C. Lee*. “Highly efficient blue InGaN nanoscale light-emitting diodes.” *Nature* **608**, 56–61 (2022).
12. K. Kang, J. Byun, M. Jeon, G. Jo, Y. K. Baek, J. Lee, H. Jeon. “Effect of sintering time on electronic properties of strontium hexaferrite.” *Ceramics International* **48**(9), 12476–12482 (2022).
13. K. Lee, J. Byun, K. Park, S. Kang, M.-S. Song, J. Park, J. Lee, S. Chae*. “Giant Tunneling Electroresistance in Epitaxial Ferroelectric Ultrathin Films Directly Integrated on Si.” *Applied Materials Today* **26**, 101308 (2022).
14. J. Lee, M. S. Song, W.-S. Jang, J. Byun, H. Lee, M. H. Park, J. Lee, Y.-M. Kim*, S. C. Chae*, T. Choi*. “Modulating the ferroelectricity of hafnium zirconium oxide ultrathin films via interface engineering to control the oxygen vacancy distribution.” *Advanced Materials Interfaces* **9**(7), 2101647 (2022).
15. Y. Ha, J. Byun, J. Lee, S. Lee. “Design Principles for the Enhanced Transparency Range of Correlated Transparent Conductors.” *Laser & Photonics Reviews* **15**(5), 2000444 (2021).
16. Y.-I. Kim[¶], M. Jeong[¶], **J. Byun[¶]**, S.-H. Yang, W. Choi, W.-S. Jang, J. Jang, K. Lee, Y. Kim, J. Lee, E. Lee, Y.-M. Kim*. “Atomic-scale identification of invisible cation vacancies at an oxide homointerface.” *Materials Today Physics* **16**, 100302 (2021).

17. J. Jo[¶], J. Byun, J. Lee, D. Choe, I. Oh, J. Park, M.-J. Jin, J. Lee*, J.-W. Yoo*. “Emergence of Multispininterface and Antiferromagnetic Molecular Exchange Bias via Molecular Stacking on a Ferromagnetic Film.” *Advanced Functional Materials* **30**(11), 1908499 (2020).
18. J. Jo[¶], J. Byun, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J. Lee*, J.-W. Yoo*. “Molecular tunability of magnetic exchange bias and asymmetrical magnetotransport in metalloporphyrin/Co hybrid bilayers.” *ACS Nano* **13**(1), 894–903 (2019).
19. S.-J. Kwon, B.-M. Jung, T. Kim, J. Byun, J. Lee, S.-B. Lee, U.-H. Choi*. “Influence of Al₂O₃ Nanowires on Ion Transport in Nanocomposite Solid Polymer Electrolytes.” *Macromolecules* **51**(24), 10194–10201 (2018).
20. T. T. Ly, G. Duvjir, T. Min, J. Byun, T. Kim, M. M. Saad, N. T. M. Hai, S. Cho, J. Lee*, J. Kim*. “Atomistic study of the alloying behavior of crystalline SnSe_{1-x}S_x.” *Physical Chemistry Chemical Physics* **19**(32), 21648–21654 (2017).

2026 items (Google Scholar-confirmed)

21. **J. Byun** (first author), J.-G. Min, S. Kim, J. Son, J. Kim, Y. Lee, S. Kim. “Retention-Resilient ECRAM–PIM Training System With Device-Algorithm-Controller Co-Design.” *IEEE Transactions on Circuits and Systems I* (2026).
22. H. Kim, J. Byun, J. Park, K. Lee, D. Ha, S. Kim. “A Novel Indium Gallium Zinc Oxide Channel-Based Electrochemical Field Effect Transistor for Physical Reservoir Computing.” *Advanced Intelligent Systems* **8**(1), 2500285 (2026).

Patent

- M. H. Sheen, Y. H. Ko, D. U. Kim, N. R. Ahn, C. H. Lee, D. H. Kim, R. Kim, I. P. Kim, et al., J. Byun. “Light emitting element, method for fabricating the same and display device.” US Patent App. 18/093,959 (2023).

Preprint

- **J. Byun** (first author), K. Lee, M. Jeong, E. Lee, J. Bang, H. Kim, G. H. Gu*, S. H. Oh*. “Imaging 3D polarization dynamics via deep learning 4D-STEM.” *arXiv:2506.06598* (2025).

CONFERENCE PRESENTATIONS

Talks

1. J. Byun, Z. Wang, S. Oh, J. Lee. (2021). “Novel Stabilization Mechanism on Polar Oxide Surface.” 2021 Materials Research Society Fall Meeting.
2. J. Byun, Z. Wang, S. Oh, J. Lee. (2021). “Novel Stabilization Mechanism on Polar Oxide Surface.” 2021 Korean Physics Society Fall Meeting.
3. J. Byun, J. Lee. (2021). “Giant Nonlinear Tunneling Current in HfO₂-based Anti-Ferroelectric Tunnel Junction.” Electronic Materials and Applications Conference 2021.
4. J. Byun, J. Lee, T. Choi, J. Lee. (2021). “Interface engineered TER optimization in HfO₂-based FTJ.” 2021 Korean Physics Society Spring Meeting.
5. J. Byun, J. Lee, T. Choi, J. Lee. (2020). “DFT investigation on structural stability on the ferroelectric HfO₂ on the electrode.” 2020 Korean Physics Society Fall Meeting.
6. J. Byun, T. Min, J. Lee. (2020). “Nonlinear current-voltage behavior in HfO₂-based ferroelectric domain-wall tunnel junction.” 2020 Korean Ceramic Society Spring Meeting.
7. J. Byun, T. Min, J. Lee. (2020). “Giant Nonlinear Tunneling Current in HfO₂-based Anti-Ferroelectric Tunnel Junction.” American Physical Society March Meeting 2020.
8. J. Byun, T. Min, J. Lee. (2019). “First-principles studies of the effects of oxygen vacancies on the HfO₂-based ferroelectric tunnel junction.” 2019 Materials Research Society Fall Meeting.

9. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the HfO₂-based ferroelectric tunnel junction." 5th International Conference on Advanced Electromaterials.
10. J. Byun, T. Min, J. Lee. (2019). "Design and modeling of optimal HfO₂-based ferroelectric tunneling junction." 2019 Korean Physics Society Fall Meeting.
11. J. Byun, T. Min, J. Lee. (2019). "Design and modeling of optimal HfO₂-based ferroelectric tunneling junction." Asia-Pacific PFM 2019 Conference.
12. J. Byun, T. Min, J. Lee. (2019). "Complex bandstructure calculation for the ferroelectric tunneling junction current." 2019 Korean Physics Society Spring Meeting.
13. J. Byun, T. Min, J. Lee. (2018). "First-principles studies of the effects of effective mass on the HfO₂-based ferroelectric tunnel junction." 2018 Korean Physics Society Fall Meeting.
14. J. Byun, T. Min, J. Lee. (2018). "Novel high coercivity Fe_{16-x}Al_xN₂ alloy." ICAUMS 2018.
15. J. Byun, T. Min, J. Lee. (2018). "High coercivity Fe_{16-x}Al_xN₂ alloy design." Korean Magnetics Society 2018 Spring Conference.
16. J. Byun, T. Min, J. Lee. (2018). "Novel high coercivity Fe_{16-x}Al_xN₂ alloy." American Physical Society March Meeting 2018.
17. J. Byun, T. Min, J. Lee. (2017). "First-principles modeling of electron tunneling through ferroelectric tunnel junction." 2017 KPS Fall Meeting.
18. J. Byun, T. Min, J. Lee. (2017). "Design of Fe_{16-x}Al_xN₂ with high-coercivity." 2017 KPS Spring Meeting.

Posters

1. J. Byun, J. Jo, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J.-W. Yoo, J. Lee. (2019). "First principles study of magnetic exchange interaction in Metalloporphyrin/Co interface." 3rd Workshop on Functional Materials Science.
2. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the HfO₂-based ferroelectric tunnel junction." 15th KIAS Electronic Structure Calculation Workshop.
3. J. Byun, T. Min, J. Lee. (2019). "Ab initio study of the role of effective mass in ferroelectric tunneling junction." 15th Dielectric Union Symposium.
4. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the effects of oxygen vacancy and effective mass on the HfO₂-based ferroelectric tunnel junction." 10th APCTP Workshop on Multiferroics.
5. J. Byun, T. Min, J. Lee. (2018). "First-principles studies of the effects of oxygen vacancies on the HfO₂-based ferroelectric tunnel junction." 12th Japan-Korea Conference on Ferroelectrics.

TECHNICAL SKILLS

DFT/MD codes: VASP, Quantum Espresso, CP2K, Wannier90, Phonopy, GULP · **ML:** PyTorch · **Programming:** Python, C++ (Arduino) · **Scientific computing:** Linux server management, SGE, Git

OTHER

- Military service (completed)
- 2021 — President of the (Physics Dept.) Student Council
- 2018 — KIAS CAC Summer School on Scientific Computing and Machine Learning
- 2017 — Python study manager for graduate students